

ASSESSMENT TOOL-2

CO2-AT2-Debugging

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1. Find the bugs, include the missing lines in the following snippet.

```
int linearSearch(int arr[], int n, int key) {  
    for(int i = 0; i &lt;= n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

Answer:

```
1  #include <stdio.h>  
2  
3  int linearSearch(int arr[], int n, int key) {  
4      for (int i = 0; i < n; i++) {  
5          if (arr[i] == key)  
6              return i;  
7      }  
8      return -1;  
9  }  
10  
11 int main() {  
12     int arr[] = {10, 20, 30, 40, 50};  
13     int n = 5, key = 30;  
14  
15     int result = linearSearch(arr, n, key);  
16  
17     if (result != -1)  
18         printf("Element found at index %d\n", result);  
19     else  
20         printf("Element not found\n");  
21  
22     return 0;  
23 }
```

Output:

```
Element found at index 2
```

2. Buggy Code in an array

```
#include <stdio.h>

int main() {
    int arr[5] = {10,20,30,40,50};

    for(int i=0;i<=5;i++)
        printf("%d ",arr[i]);

    return 0;
}
```

Expected output:

```
10 20 30 40 50
```

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[5] = {10, 20, 30, 40, 50};
5
6      for(int i = 0; i < 5; i++) {
7          printf("%d ", arr[i]);
8      }
9
10     return 0;
11 }
```

Output:

```
10 20 30 40 50
```

3. Identify an errors in the following snippet

```
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;

    while(low < high)
    { int mid = (low + high) / 2;
```

```
if(arr[mid] == key) return mid;

else if(arr[mid] < key)

low = mid;

else high = mid;

}

return -1;

}
```

Answer:

```
1  #include <stdio.h>
2
3  int binarySearch(int arr[], int n, int key) {
4      int low = 0, high = n - 1;
5
6      while(low <= high) {
7          int mid = (low + high) / 2;
8
9          if(arr[mid] == key)
10             return mid;
11         else if(arr[mid] < key)
12             low = mid + 1;
13         else
14             high = mid - 1;
15     }
16
17     return -1;
18 }
19
20 int main() {
21     int arr[] = {10, 20, 30, 40, 50};
22     int n = 5, key = 30;
23
24     int result = binarySearch(arr, n, key);
25
26     if(result != -1)
27         printf("Element found at index %d\n", result);
28     else
29         printf("Element not found\n");
30
31     return 0;
32 }
```

Output:

```
Element found at index 2
```